



CONFIGURED PRIMARY TUMOR	CUPPA CANCER TYPE	QC
cancer (DOID 162)	Skin: Melanoma (100%)	PASS

Purity: 99% (97%-100%)

Ploidy: 3.10 (3.05-3.15)

Somatic variant drivers: 6 (BRAF, CDKN2A, TERT)

Germline variant drivers: None

Somatic copy number drivers: 1 (PTEN)

Germline copy number drivers: None

Somatic disruption drivers: None

Germline disruption drivers: None

Fusion drivers: None

Viral presence: None

Whole genome duplicated: Yes

Microsatellite indels per Mb: 0.1 (Stable)

Tumor mutations per Mb: 13.7 (High)

Tumor mutational load: 187 (High)

HR deficiency score: 0.0 (Proficient)

DPYD status: *1 HOM (Normal Function)

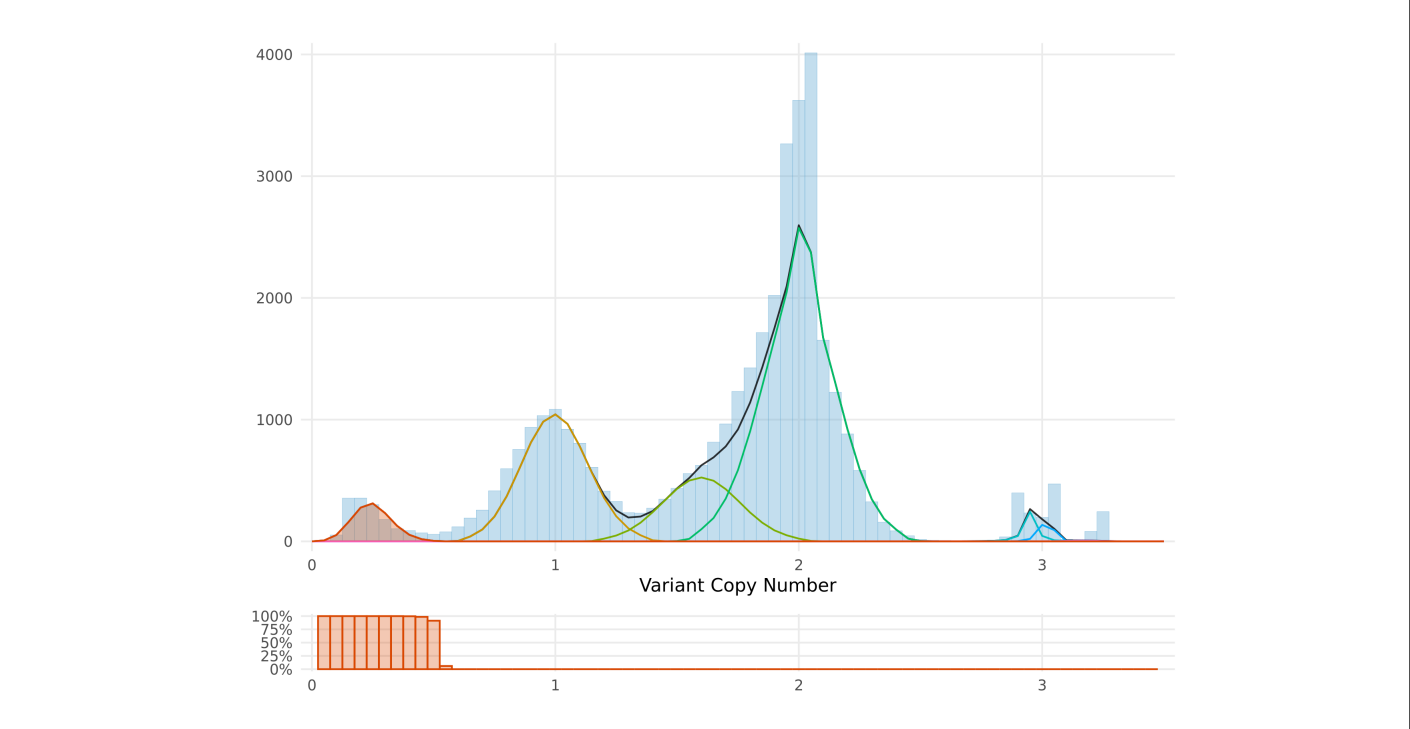
UGT1A1 status: NA

Number of SVs: 92 (Pan 0.25)

Max complex cluster size: 8

Telomeric SGLs: 0

Number of LINE insertions: 0





Somatic Findings

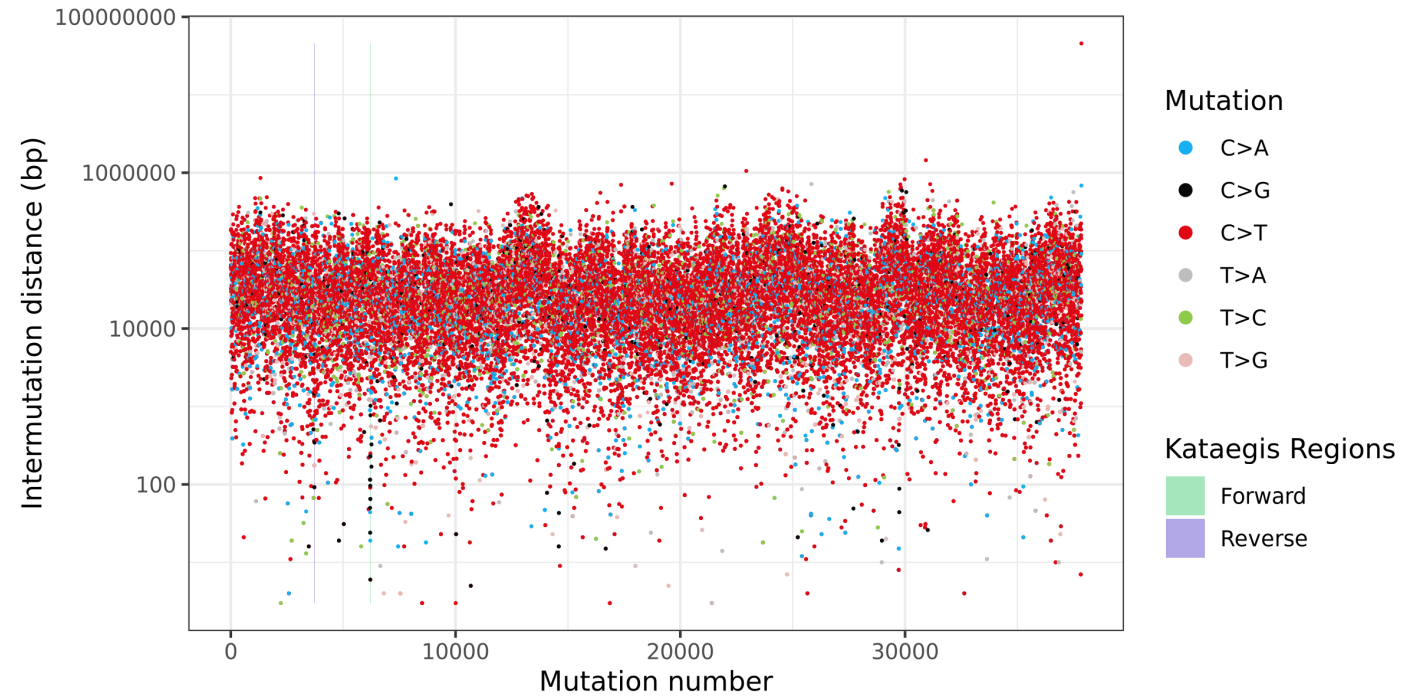
Driver variants (7)

VARIANT	VCN	CN	MACN	BIALLELIC	HOTSPOT	DL	CL	PHASE ID	RNA DEPTH
BRAF p.V600E	4.1	6.0	2.0	2%	Yes	100%	100%		NA
CDKN2A p.A68fs	2.0	2.0	0.0	100%	Near	100%	100%		NA
CDKN2A (alt) p.G83fs	2.0	2.0	0.0	100%	Near	100%	100%		NA
TERT c.-125_-124delCCinsTT	1.7	2.0	0.0	92%	Yes	100%	100%	4295	NA
HDAC2 p.R409*	0.9	2.9	1.0	2%	No	34%	100%		NA
SF3B1 p.P718L	2.0	3.0	1.0	2%	No	15%	100%		NA
TP63 p.M499I	1.8	4.0	2.0	2%	No	0%	100%		NA

Other potentially relevant variants (3)

VARIANT	VCN	CN	MACN	BIALLELIC	HOTSPOT	DL	CL	PHASE ID	RNA DEPTH
ALB c.1059-46dupT	1.8	3.9	1.9	2%	No		100%		NA
MED12 c.3210-6C>T	2.0	2.0	0.0	100%	No		100%		NA
STK19 upstream	2.0	3.8	1.8	2%	Yes		100%		NA

Kataegis plot



Driver amplifications and homozygous deletions (1)

LOCATION	GENE	TYPE	CN	TPM	PERC (TYPE)	FC (TYPE)	PERC (DB)	FC (DB)
chr10q23.31	PTEN	partial del	0.0	NA	NA	NA	NA	NA



Potentially interesting near-driver amplifications (0)

NONE

Other regions with amplifications (1)

LOCATION	GENE	TYPE	CN	TPM	PERC (TYPE)	FC (TYPE)	PERC (DB)	FC (DB)
chr3p24.2	CFL1P7	full gain	12.1	NA	NA	NA	NA	NA

Regions with deletions in genes in other autosomal regions (1)

LOCATION	GENE	TYPE	CN	TPM	PERC (TYPE)	FC (TYPE)	PERC (DB)	FC (DB)
chr16q21	CNOT1	partial del	0.0	NA	NA	NA	NA	NA

Potentially interesting chromosomal rearrangements

CHROMOSOMAL REARRANGEMENT	DETECTED?
1q trisomy	No
1p19q co-deletion	No

Driver fusions (0)

NONE

Other potentially interesting fusions (0)

NONE

Potentially interesting in-frame fusions in case no high drivers detected

High driver likelihood events are detected in this sample, therefore this section is empty

Driver viruses (0)

NONE

Other viral presence (0)

NONE

Homozygous disruptions (0)

NONE

Driver gene disruptions (2)

LOCATION	GENE	RANGE	TYPE	CLUSTER ID	JUNCTION CN	UNDISRUPTED CN
chr10q23.31	PTEN	Intron 5 Upstream	DEL	45	2.0	0.0
chr10q23.31	PTEN	Intron 6 Downstream	DEL	45	2.0	0.0



Other potentially interesting gene disruptions (0)

NONE

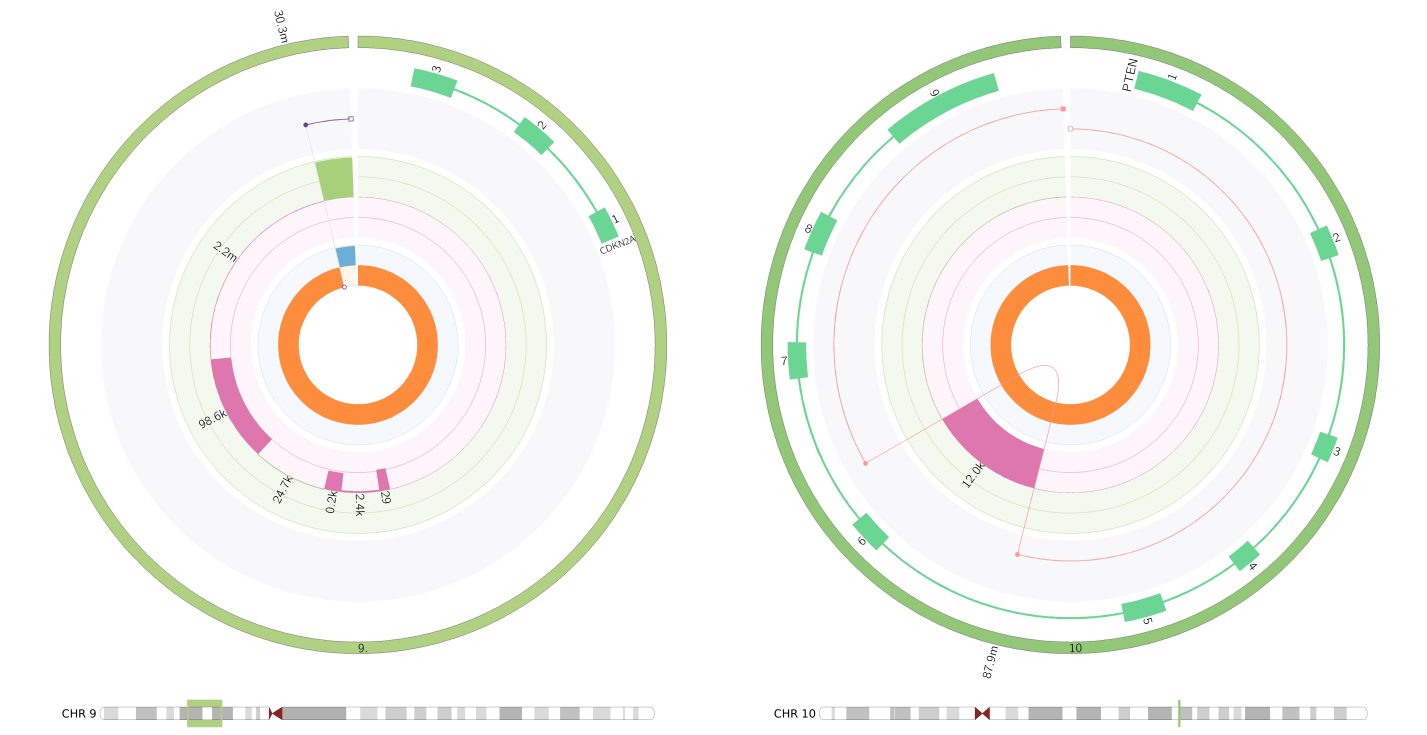
Potentially interesting LOH events (0)

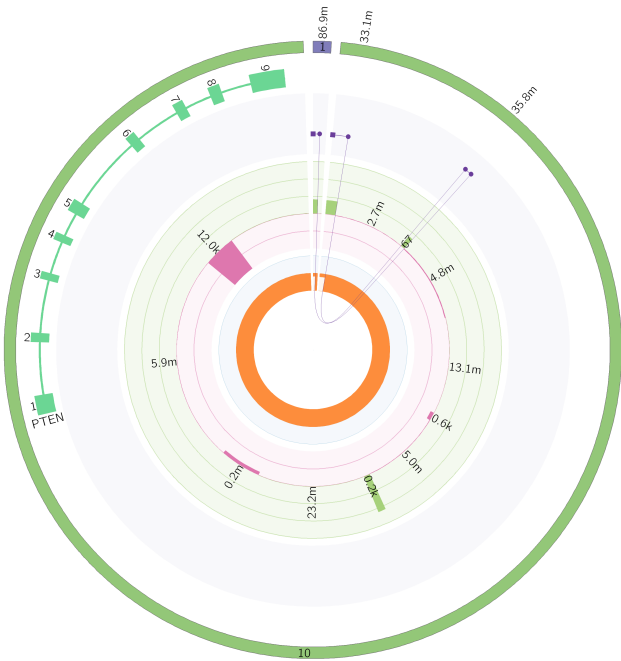
NONE

Signature allocations (11)

SIGNATURE	ETIOLOGY	ALLOCATION	PERCENT
Sig7	Ultraviolet light exposure	23193.2	61%
Sig2	APOBEC activity	3881.0	10%
Sig4	Tobacco smoking	2330.8	6%
Sig18	Reactive oxygen species	2013.2	5%
Sig8	Unknown	1805.4	5%
Sig11	Temozolomide treatment	1455.8	4%
Sig12	Unknown	1229.8	3%
Sig17	Unknown	1121.2	3%
Sig28	Unknown	570.8	2%
Sig24	Aflatoxin exposure	246.8	1%
MISALLOC	-	4846.1	13%

Structural driver plots (3)







Germline Findings

Driver variants (0)

NONE

Other potentially relevant variants (4)

VARIANT	VCN	CN	MACN	RNA DEPTH	BIALLELIC	HOTSPOT	GENOTYPE
CYP2D6 c.506-1G>A splice	1.9	4.0	2.0	NA	No	Yes	HET
CYP3A4 c.522-191C>T	2.4	4.0	2.0	NA	No	Yes	HET
CYP3A4 upstream	4.0	4.0	2.0	NA	Yes	Yes	HOM
CYP3A5 c.219-237A>G	4.0	4.0	2.0	NA	Yes	Yes	HOM

Potentially pathogenic germline deletions (0)

NONE

Potentially pathogenic germline LOH events (0)

NONE

Potentially pathogenic germline homozygous disruptions (0)

NONE

Potentially pathogenic germline gene disruptions (0)

NONE

Genes with missed variant likelihood > 1% (27)

GENE	MVLH	GENE	MVLH	GENE	MVLH	GENE	MVLH
AMER1	3.8%	AR	2.6%	ARAF	2.1%	ATP2B3	2.0%
ATRX	3.1%	BCOR	2.2%	BTK	3.2%	CASZ1	3.5%
DDX3X	2.7%	GATA1	1.9%	GNAS	1.9%	GTF2I	26.9%
KDM5C	1.7%	KDM6A	3.3%	KRTAP5-5	1.4%	MED12	3.3%
OR11H1	2.5%	PHF6	3.9%	PMS2	4.6%	RBM10	2.0%
RPS6KA3	3.7%	SMARCA2	1.4%	STAG2	2.5%	USP9X	2.5%
ZFX	1.5%	ZMYM3	3.0%	ZRSR2	5.0%		

Germline CN aberrations (0)

NONE

Pharmacogenetics (1)

GENE	HAPLOTYPE	GENOTYPE	FUNCTION	LINKED DRUGS	SOURCE
DPYD	*1	HOM	Normal Function	5-Fluorouracil;Capecitabine;Tegafur	PHARMGKB



Immunology

HLA QC

QC Status: PASS

HLA Alleles (6)

ALLELE	REF FRAGS	TUMOR FRAGS	RNA FRAGS	TUMOR CN	SOMATIC #MUTATIONS
A*01:01	210	1607	NA	2.0	NONE
A*01:01	210	1607	NA	1.8	NONE
B*08:01	219	755	NA	1.8	NONE
B*40:02	197	727	NA	2.0	NONE
C*03:04	213	769	NA	2.0	NONE
C*07:01	218	766	NA	1.8	NONE

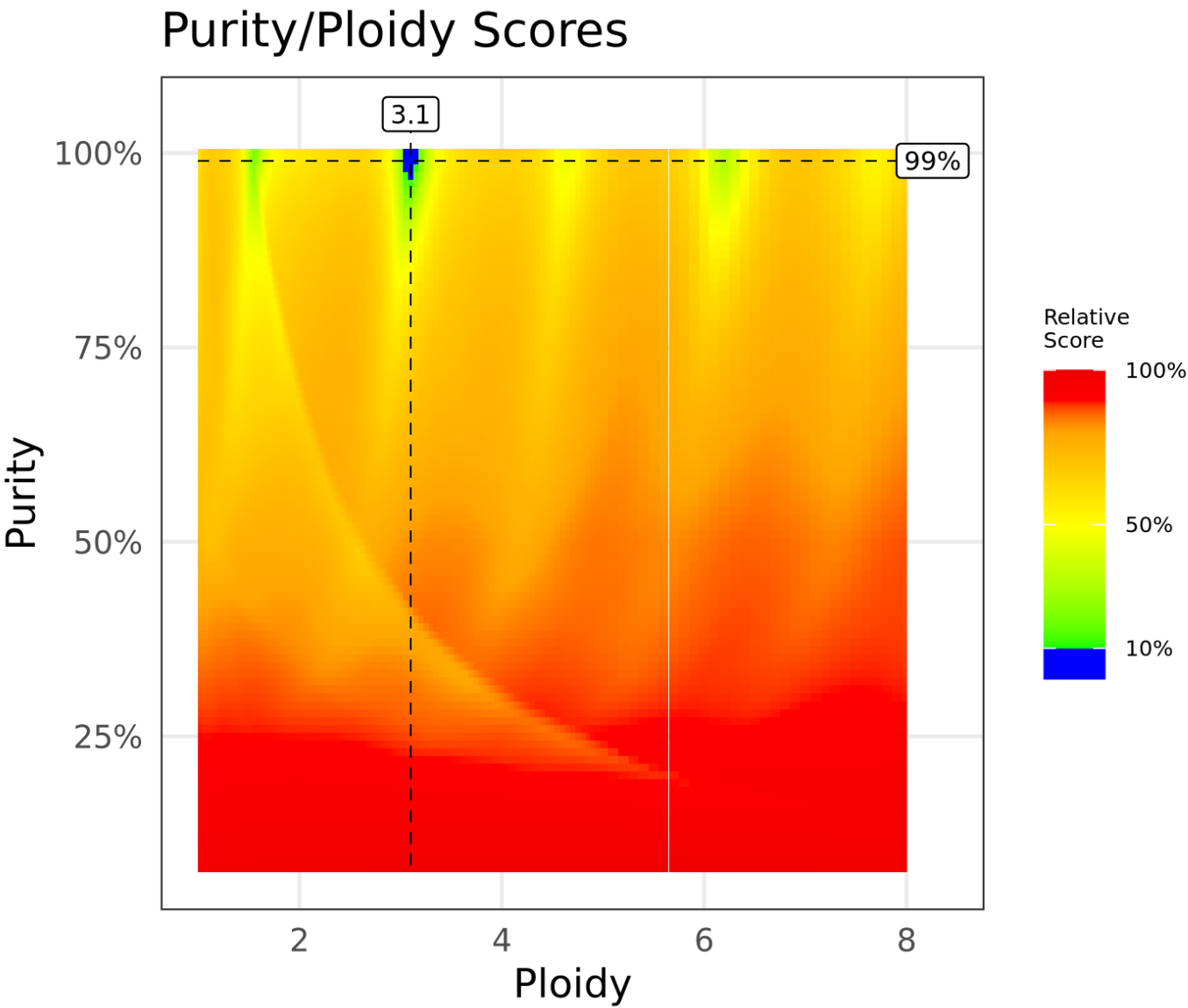
Genetic Immune Escape

ESCAPE MECHANISM	DETECTED?
HLA-1 loss-of-function	No
Antigen presentation pathway inactivation	No
IFN gamma pathway inactivation	No
(Potential) PD-L1 overexpression	No
CD58 inactivation	No
Epigenetics driven immune escape via SETDB1	No



Quality Control

QC	REF GENOME	FIT METHOD	MEAN DEPTH	CONTAMINATION	UNS. SEGMENTS (%)	DELETED GENES
PASS	V38	NORMAL	108	0%	0 (0%)	4



Flagstats

	UNIQUE RC	SECONDARY RC	SUPPLEMENTARY RC	MAPPED PROPORTION
Ref Sample	702786970	0	823759	100%
Tumor Sample	2550957378	0	2993259	100%



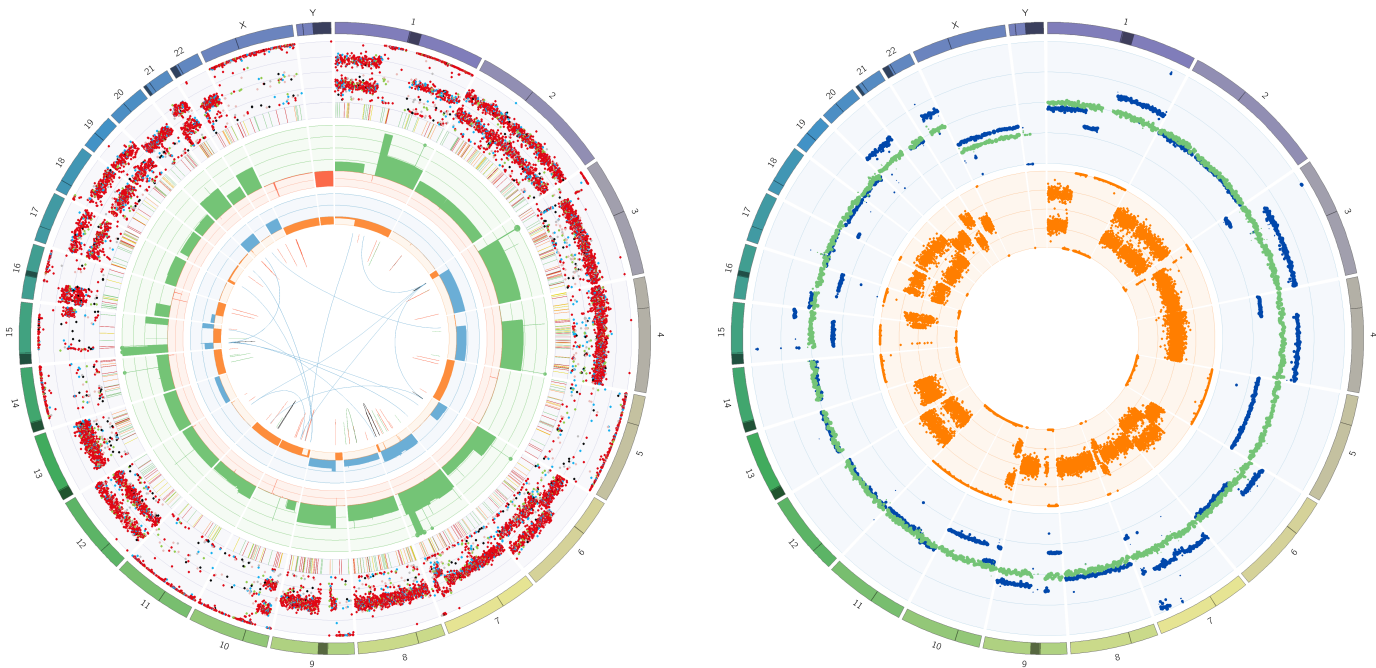
Coverage Stats

	MEAN COVERAGE	SD COVERAGE	MEDIAN COVERAGE	MAD COVERAGE
Ref Sample	30.6	9.6	32	5
Tumor Sample	105.7	36.9	108	24

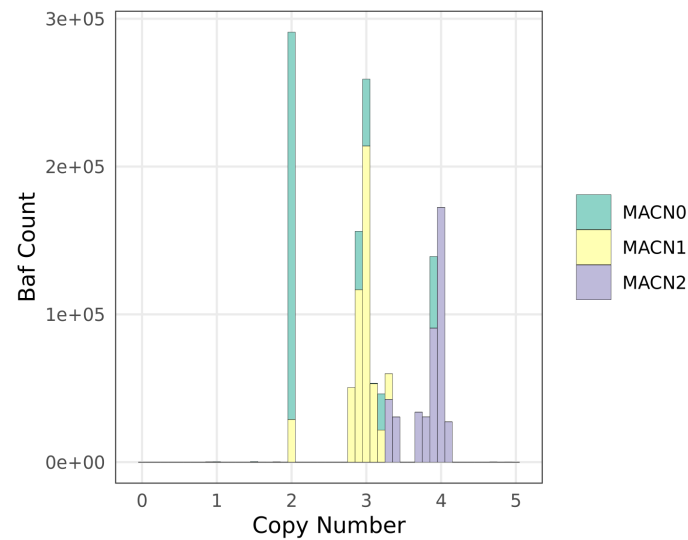
Excluded Percentages

	ADAPTER	BASEQ	CAPPED	DUPE	MAPQ	OVERLAP	UNPAIRED	TOTAL
Ref Sample	0%	0%	0%	11%	3%	0%	0%	15%
Tumor Sample	0%	0%	0%	14%	3%	1%	0%	19%

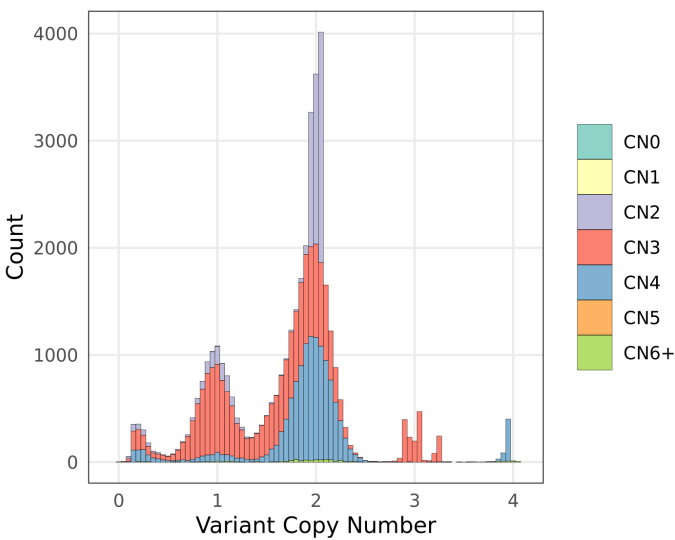
QC plots



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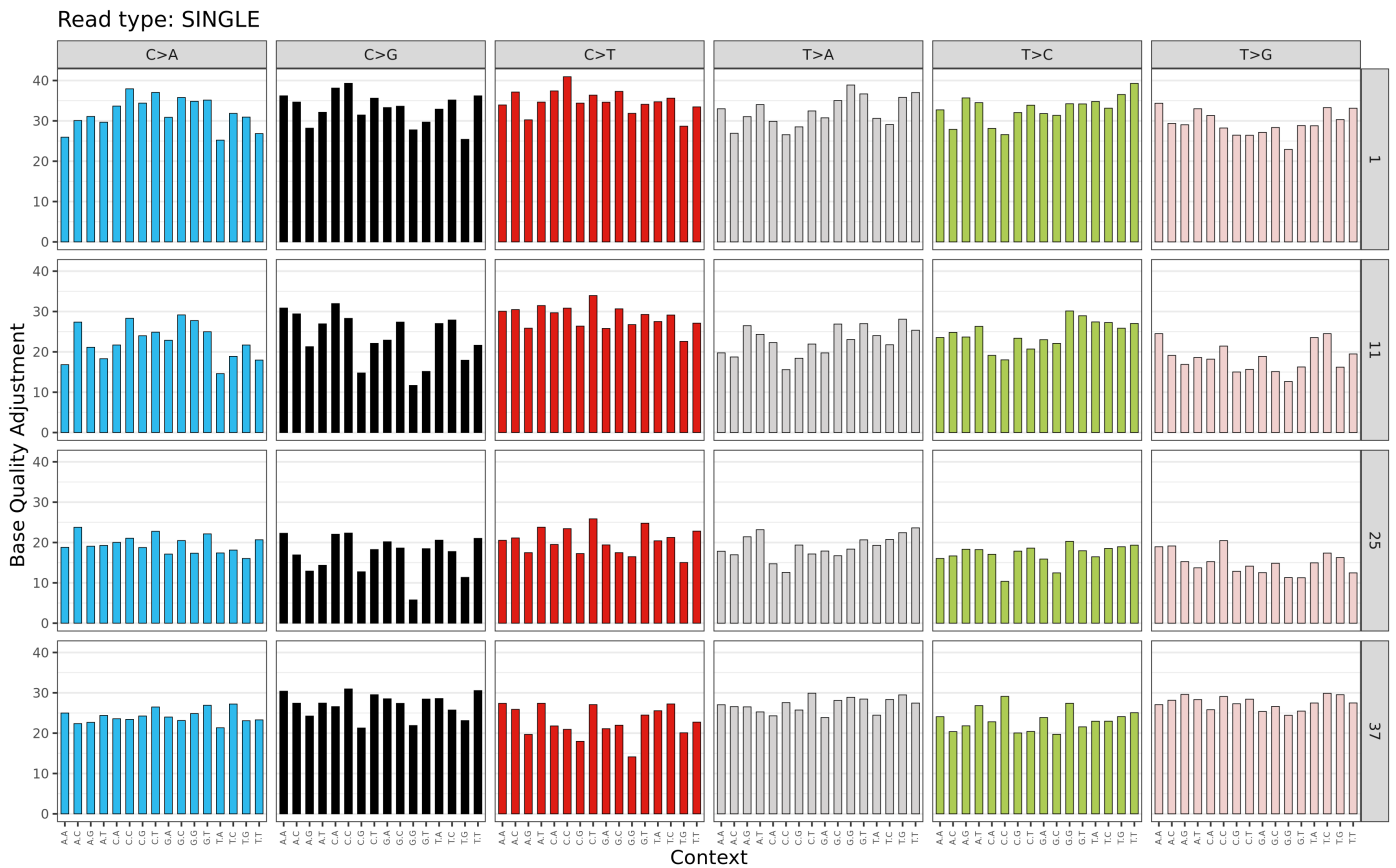
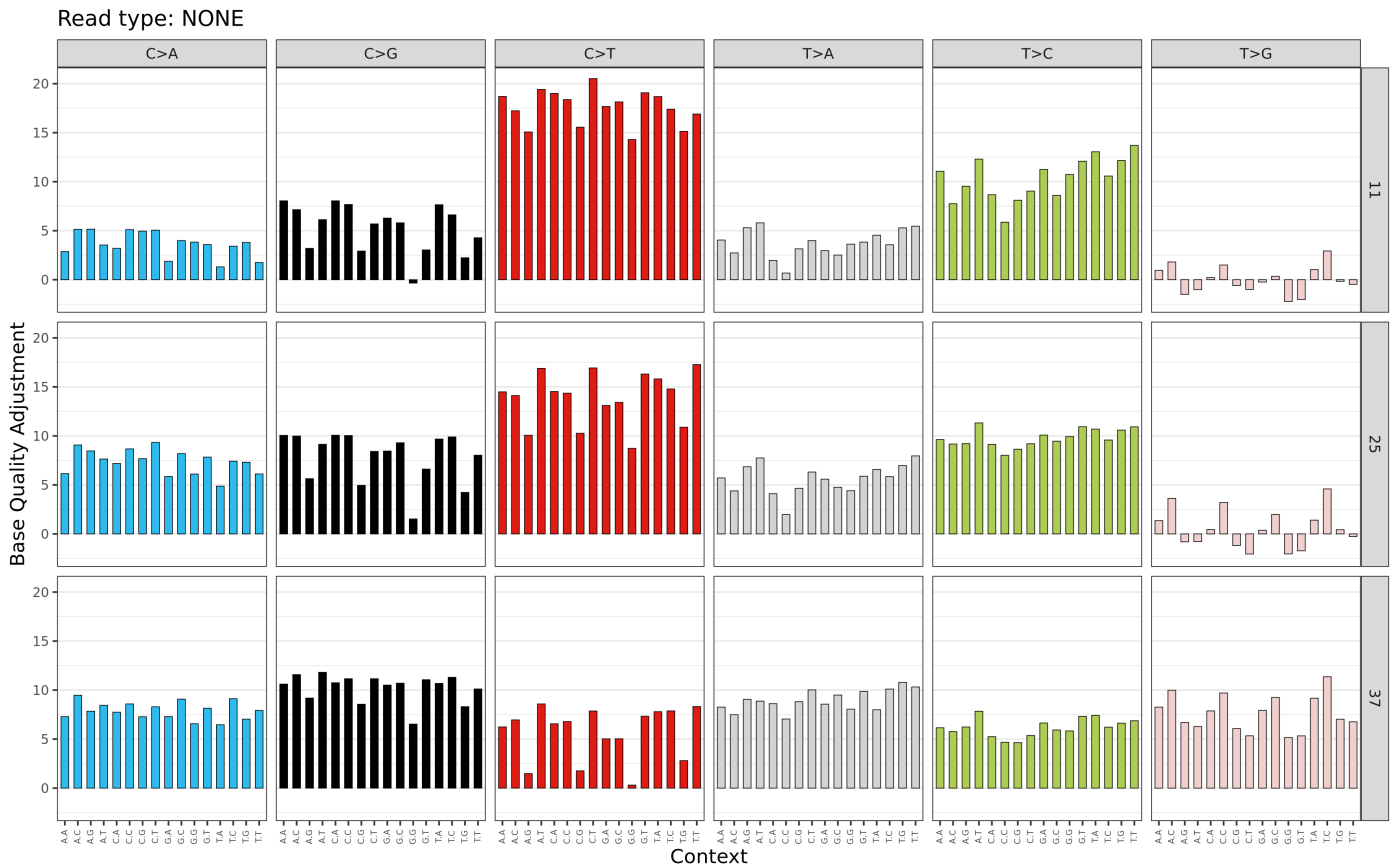


Somatic Variant Copy Number PDF



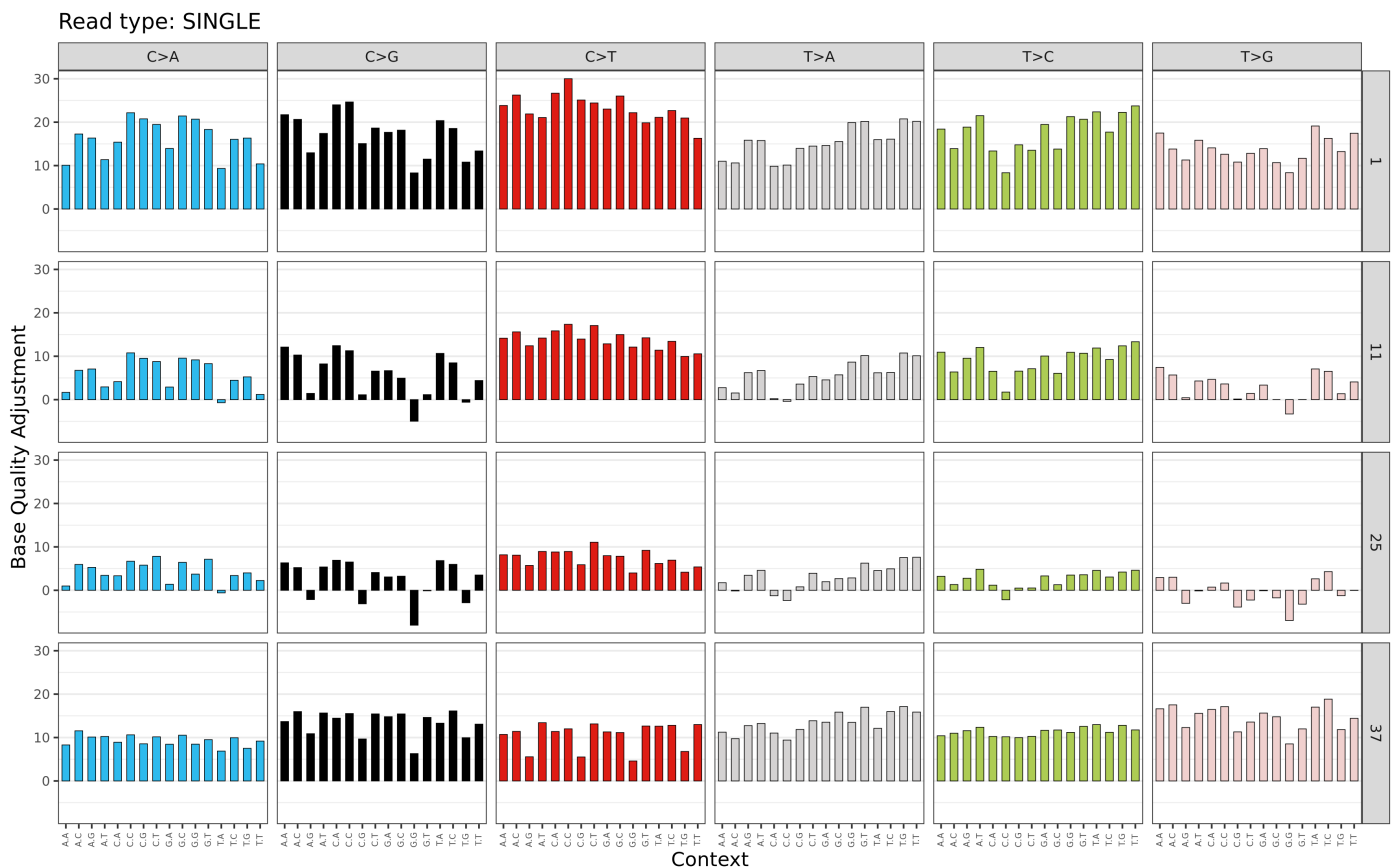
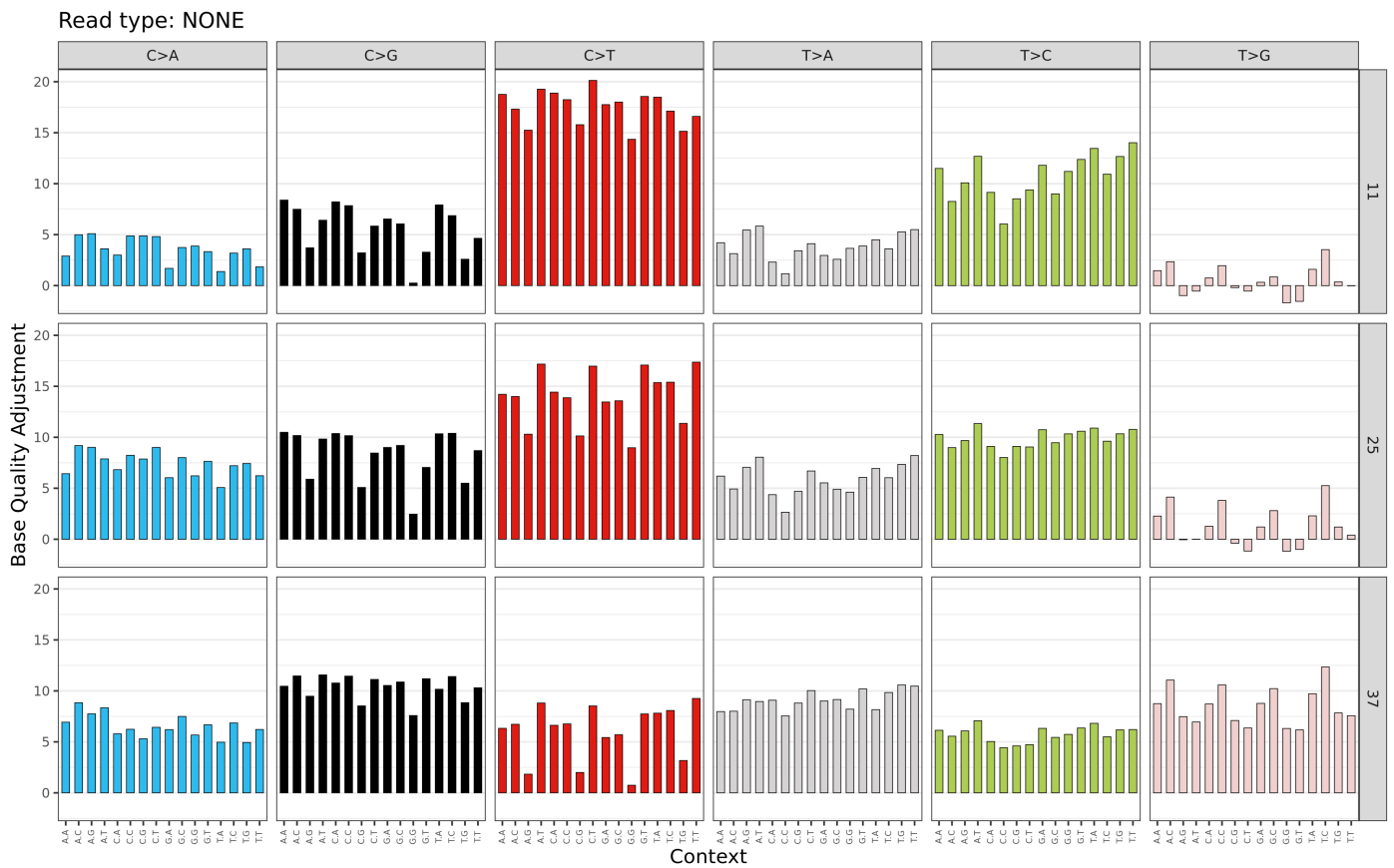


Reference Sample BQR plot





Tumor Sample BQR plot





Tumor Detection Statistics

Tumor maximum diploid proportion	0.02
Number of hotspot mutations	3
Number of hotspot structural variants	0
Sum of small variant allele read counts	2288506
Sum of structural variant tumor fragment counts (excluding single breakends)	6526
Sum of B-allele frequency points in germline diploid regions with tumor ratio < 0.8 OR > 1.2	675339